

# Adam Bouafia

Amsterdam, Netherlands | adam.bouafia@hotmail.com | 06 17245330 | adam-bouafia.github.io  
linkedin.com/in/adam-bouafia | github.com/adam-bouafia

## Summary

---

DevOps and Platform Engineer with an MSc in Computer Science. Hands-on experience with Azure, Kubernetes, Terraform and CI/CD pipelines, including production-grade cloud infrastructure built entirely with Infrastructure as Code. Currently completing the Linux Foundation Advanced Cloud Engineer program and preparing for the CKA. Focused on automation, observability and green computing.

## Experience

---

**DevOps Research Engineer**, Vrije Universiteit Amsterdam – Amsterdam, Netherlands Mar 2025 – Jan 2026

- Built semantic log compression system processing 2.4M+ logs with 22.33x compression ratio
- Implemented Lazy Reconstruction query system achieving 21-111ms latency
- Developed containerized pipeline with Docker and automated CI/CD via GitHub Actions

**DevOps Engineer**, Vivieb – Pescara, Italy Nov 2023 – Feb 2024

- Optimized Kubernetes cluster energy efficiency using Kubegreen
- Built monitoring dashboards with Grafana and InfluxDB
- Implemented automated deployment pipelines

**Software Developer**, Euromed Organization – Paris, France July 2022 – Dec 2022

- Developed Dhayen mobile security app for national awareness campaign
- Implemented CI/CD pipeline for mobile app deployment

## Professional Development

---

**The Linux Foundation - Advanced Cloud Engineer IT Professional Program** Mar 2026 – present

- Eight-course cloud native track: Containers (LFS253), Kubernetes (LFS258), Prometheus (LFS241), Fluentd and Fluent Bit (LFS242), Service Mesh (LFS243), Helm (LFS244)
- Built and operated a three-node kubeadm cluster on libvirt/KVM: cluster build and upgrade, etcd backup and restore, control plane recovery, RBAC, CNI networking and live failure troubleshooting
- Preparing for the CKA (Certified Kubernetes Administrator) exam

**Microsoft Azure Certification Path** In progress

- Preparing for AZ-104 (Azure Administrator Associate), followed by AZ-400 (Designing and Implementing Microsoft DevOps Solutions) toward the DevOps Engineer Expert credential

## Certifications

---

**CKA (Certified Kubernetes Administrator)** Sept 2026

## Education

---

**Vrije Universiteit Amsterdam / Università dell'Aquila**, MSc in Computer Science (Double Degree)

- Specialization: Software Engineering & Green IT

## Projects

---

### Azure Landing Zone

- Production Azure infrastructure with hub-spoke topology using 100% Bicep IaC
- Azure Firewall, Bastion, NSG, Key Vault with RBAC
- Multi-stage CI/CD pipeline (Azure DevOps) with validation, what-if, and approval gates
- Automated cost optimization scripts saving 70% on dev infrastructure

### Greenmining

- Python library published on PyPI for Green IT research
- 124 sustainability patterns from Green Software Foundation
- Multi-backend energy measurement (RAPL, CodeCarbon)
- GitHub Actions CI/CD with automated testing and deployment

### LCT - LLM Comparative Tool

- Open-source framework evaluating 500K+ HuggingFace models
- Dockerized deployment with CI/CD pipeline
- Real-time energy profiling and carbon footprint analytics

### CO2 Consumption Monitor

- GNOME Shell extension published on GNOME Store
- Real-time CO2 emissions tracking with ElectricityMaps API integration
- Per-application monitoring and historical data export

### IoT Monitoring System

- Real-time monitoring using MQTT, Node-RED, InfluxDB, Grafana
- Automated alerts via Telegram integration
- Linux-based deployment on Raspberry Pi

## Skills

---

**Cloud Platforms:** Azure

**Containerization & Orchestration:** Kubernetes (kubeadm, RBAC, etcd operations), Docker, Helm, Knative

**Infrastructure as Code:** Terraform, Bicep, Ansible

**CI/CD & GitOps:** Azure DevOps, GitHub Actions, GitLab CI, Jenkins, ArgoCD

**Operating Systems:** Linux (Fedora, Ubuntu, RHEL), Windows

**Scripting & Programming:** Python, Bash, Java, JavaScript, SQL

**Monitoring & Logging:** Prometheus, Grafana, InfluxDB, ELK Stack, Fluentd, Fluent Bit

**Version Control:** Git, GitHub, GitLab

## Languages

---

**English:** Native

**Dutch:** Beginner

**Arabic:** Native

**French:** Proficient

**Italian:** Beginner